Description

Image





Caption

1. Nickel used in almost pure form for coinage. © ANSYS, Inc. 2. AA battery. © Sbn1984 at en.wikipedia - Public domain

The material

The US 5¢ piece is, as its name suggests, almost pure nickel. Look at one - much can be learned from it: that nickel is a ductile, silvery metal, easily formed by stamping (or rolling or forging), hard when worked, and with good resistance to corrosion. Not bad for 5¢. Nickel has a high melting point (1450 C) and is one of the few elements that are ferro-magnetic (with Fe and Co). You don't see pure nickel very often, but its alloys are everywhere. Alloyed with copper it is widely used for coinage (the Euro, the "silver" dollar, the British 50 p piece). As Nichrome, a Ni-Cr alloy, it forms the heating elements of electric fires, toasters and hair dryers. Alloyed with iron and chromium it becomes stainless steel, familiar in every kitchen. And most exotic of all are the range of nickel-based materials known as "superalloys" because of their exceptional combination of high temperature strength and corrosion resistance. This record is for pure nickel. There are separate records for stainless steels, nickel-chromium alloys and superalloys.

Compositional summary

99.8% Ni

General properties

Density		8.83e3	-	8.95e3	kg/m^3
Price	*	13.3	-	17.1	GBP/kg
Date first used		1751			

Mechanical properties

Young's modulus	190	-	220	GPa
Shear modulus	72	-	86	GPa
Bulk modulus	162	-	200	GPa
Poisson's ratio	0.305	-	0.315	

Yield strength (elastic limit)		70	-	900	MPa
Tensile strength		345	-	1e3	MPa
Compressive strength		70	-	1e3	MPa
Elongation		2	-	60	% strain
Hardness - Vickers		80	-	300	HV
Fatigue strength at 10^7 cycles	*	135	-	500	MPa
Fracture toughness		80	-	110	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	2e-4	-	0.0032	

Thermal properties

Melting point		1.44e3	-	1.47e3	°C
Maximum service temperature	*	240	-	370	°C
Minimum service temperature		-272	-	-271	°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity		67	-	91	W/m.°C
Specific heat capacity		452	-	460	J/kg.°C
Thermal expansion coefficient		12	-	13.5	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		8	-	10	µohm.cm

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		Yes			

Processability

Castability		3			
Formability		3	-	4	
Machinability		3			
Weldability		4	-	5	
Solder/brazability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			
Soils, alkaline (clay)		Excellent			

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Wine	Excellent

Durability: acids

Acetic acid (10%)	Acceptable
Acetic acid (glacial)	Excellent
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Acceptable
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Limited use
Nitric acid (10%)	Limited use
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Limited use
Phosphoric acid (85%)	Limited use
Sulfuric acid (10%)	Acceptable
Sulfuric acid (70%)	Acceptable

Durability: alkalis

Sodium hydroxide (10%)	Acceptable
Sodium hydroxide (60%)	Acceptable

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Acceptable
Acetone	Excellent

Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Excellent
O2 (oxygen gas)	Acceptable
Sulfur dioxide (gas)	Limited use

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	2.37e6	tonne/yr
Reserves, principal component	1.05e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 14	-	15.5	kg/kg
Embodied energy (virgin grade production)	* 196	-	216	MJ/kg
Water usage	* 222	-	245	l/kg

Material processing: energy

Metal casting, embodied energy	* 47.9	-	54.2	MJ/kg

Metal extrusion, hot, embodied energy	* 6.37	-	8.63	MJ/kg
Metal rolling and forging, embodied energy	* 3.65	-	4.83	MJ/kg
Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg
Metal powder forming, embodied energy	* 65	-	74.1	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 1.04e4	-	1.27e4	MJ/kg
Machining, coarse, embodied energy	* 15.4	-	17.3	MJ/kg
Machining, fine, embodied energy	* 17.1	-	19.6	MJ/kg
Machining, grinding, embodied energy	* 18.7	-	21.9	MJ/kg
Machining, laser, CO2, embodied energy	* 1.23	-	1.51	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 3.45	-	3.92	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 0.417	-	0.566	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.238	-	0.316	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 5.3	-	6.04	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 854	-	1.04e3	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.01	-	1.13	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.12	-	1.28	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.23	-	1.43	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.102	-	0.124	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✓			
Climate change (CO2-eq), recycling	* 1.82	-	2.02	kg/kg
Embodied energy, recycling	* 13.7	-	15.1	MJ/kg
Recycle fraction in current supply	56			%
Combust for energy recovery	✗			
Landfill	✗			
Biodegrade	✗			
Toxicity rating	Slightly toxic			
A renewable resource?	✗			

Environmental notes

About 10% of the population is sensitive to nickel, causing them to react even to the nickel in stainless steel watch straps. Compounds of nickel can be more toxic; nickel carbonyl, used in the extraction of nickel, is deadly.

Supporting information

Design guidelines

Pure nickel has good electrical conductivity, thermal conductivity, and strength and corrosion resistance; nickel and its alloys are used in marine applications for heat exchanges in other structures. The allergic reaction it sometimes causes limits the use of pure nickel in consumer products, but its alloys are largely immune from this problem.

Technical notes

Most nickel is use in alloys with chromium (Nichromes, Chromels), with iron (stainless steels) and as complex super-alloys (Hastelloy, Nimonic, Udimet).

Typical uses

Coinage, battery electrodes, as an electro-plate for corrosion protection of other metals.

Links

ProcessUniverse

Producers

Reference